Joemon Joy

AI Engineer | GenAI, RAG & Forward Deployed Engineering

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PROFESSIONAL SUMMARY

AI engineer with experience contributing to LLM integrations, hybrid RAG retrieval, backend APIs, and application testing in a collaborative enterprise GenAI platform. Skilled in Python, FastAPI, PostgreSQL/pgvector, Docker, and Git-based development, with practical experience building and troubleshooting software systems.

PROFESSIONAL EXPERIENCE

CoCreate Ventures Sep 2025 – Present

Student Innovation Assistant – Research Intern (AI/ML) Bengaluru, India

ARC – Enterprise Multi-Tenant GenAI Platform

Python, FastAPI, PostgreSQL, React

- Authored **53 merged PRs** in a collaborative codebase spanning LLM integration, retrieval, evaluation, security, and application reliability.
- Integrated an **OpenRouter LLM provider** using Python HTTP clients with configurable models, bounded retries, and structured tool-call validation.
- Implemented **hybrid RAG retrieval** with PostgreSQL full-text search, pgvector embeddings, reciprocal rank fusion, relevance filtering, and tenant-access revalidation.
- Submitted code changes for **LLM retry handling**, RAG relevance checks, and security-related fixes through reviewed GitHub pull requests.
- Fixed React dashboard data-loading issues and expanded route-level smoke tests to improve frontend reliability.

SELECTED TECHNICAL PROJECTS

Logly – Distributed Log Monitoring & Detection

Go, Python, Kafka, Elasticsearch

- Developed a **Go-based Linux log collector** for SSH authentication events, structured telemetry, and Kafka-backed event delivery to a Python detection pipeline.
- Improved event-processing reliability with **acknowledgment after storage**, deterministic document IDs for Kafka-record replay, parsing safeguards, and targeted regression tests; deployment remains under validation.

Emotion Classification API

Python, PyTorch, FastAPI, Docker

- Trained a six-class **Bi-GRU classifier**, achieving **85.75% accuracy** and **81.96% macro-F1** on 2,000 held-out samples.
- Built a Dockerized FastAPI inference service and deployed it on Render (API docs).

FraiseQL – External Open-Source Contribution

Rust, Cargo, GitHub Actions

- Submitted **PR #491** fixing Rust benchmark compilation and expanding CI checks; **credited in the project contributor list** despite the PR closing unmerged after an equivalent maintainer fix.

TECHNICAL SKILLS

Languages: Python, SQL, Go; working exposure to JavaScript and Rust

AI & ML: LLM APIs, RAG, Hybrid Retrieval, pgvector, PyTorch, scikit-learn, LangChain

Backend & Data: FastAPI, REST APIs, PostgreSQL, Kafka, Elasticsearch

Engineering Tools: Docker, Git/GitHub, GitHub Actions, pytest, Linux, React

EDUCATION

Sri Siddhartha Institute of Technology, Tumakuru 2022 – 2026

B.E. in Computer Science and Engineering | Completed

CERTIFICATIONS & LEADERSHIP

Certifications: Machine Learning Specialization – DeepLearning.AI / Stanford Online; Deep Learning Specialization – DeepLearning.AI

Leadership: Secretary, IEEE SSIT Student Chapter